4.5.3 Drums/Barrels

Drums or barrels (drums) are a highly visible warning and channelizing device. They are constructed of lightweight, deformable materials. They appear to be solid and therefore command the respect of drivers.

On multilane highways, they are used to delineate opposing flows of traffic, especially where a median barrier has been removed for repaving operations.

Drums are predominantly orange, and shall have five uniformly-spaced retroreflective bands at least 100 mm (4") wide of fluorescent orange and white as shown in *Figure 4.10: Drum*.

Drums are generally at least 90 cm (36") tall, at least 45 cm (18") wide at the base, and at least 30 cm (12") wide at the top.

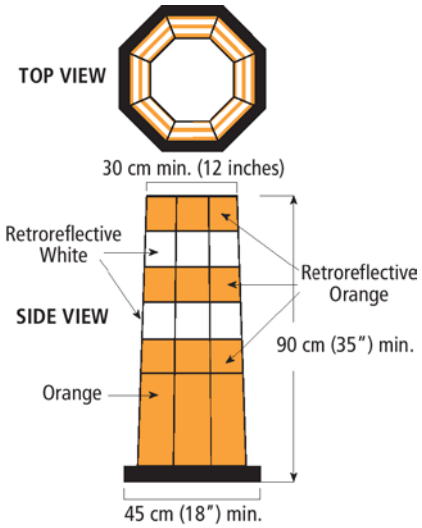


Figure 4.10: Drum

Drums are most commonly used to define leading tapers, to mark equipment areas on the side of the road, provide delineation when barrier has been removed, and in areas where additional emphasis is needed.

These principles apply when using drums:

- Drums should not be weighted with sand, water, or any other material to an extent that would make them hazardous when striking road users or workers.
- Drums used in regions susceptible to freezing should have drain holes in the bottom so that water will not accumulate and freeze.
- Ballast shall not be placed inside or on the top of a drum. If extra weight is required, sandbags or weighted rings may be added around the outside of the base.
- Flashers may be used where required, and to increase visibility (see *Section 4.9.2: Yellow Warning Lights* for more information).

4.5.4 Surface-Mounted Delineators

Surface-mounted delineators may be used on the centreline to separate opposing traffic on a two-lane, two-way roadway. They are predominantly orange, and made of a material that can be struck without causing damage to the impacting vehicle.

They are fastened directly to the pavement surface. Non-cylindrical delineators are attached to the pavement in a manner whereby they display at least a 60 mm (2.5") of width to all approaching road users.

Surface-mounted delineators shall be retroreflective or equipped with lighting devices for maximum visibility.

Retroreflectivity for 90 cm (36") or larger delineators shall be provided by a white band 100 mm (4") wide and positioned 50 mm (2") from the top of the delineator.

An additional white band 100 mm (4") wide shall be positioned approximately 150 mm (6") below the 100 mm (4") band.

Surface-mounted delineators should not be substituted for drums or barricades to mark hazards or to close roadways.

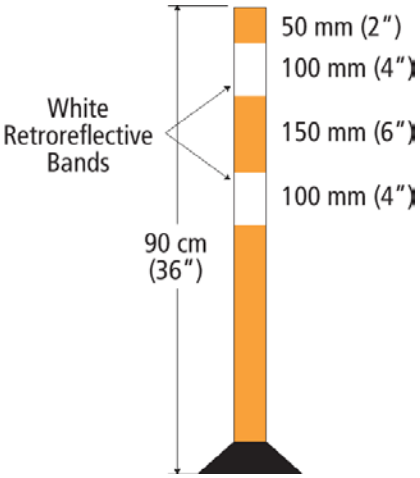


Figure 4.11:
Surface-Mounted Delineator

4.5.5 Post-Mounted Delineators

Post-mounted delineators are most commonly used in long-duration work zones to mark the edge of roadway through diversions. They are used in combination with, or to supplement other, temporary traffic control devices.

They shall be mounted on crash-worthy supports (typically square perforated tubing) so that the retroreflective surface is approximately 100 cm (39") above the nearest roadway edge. The retroreflective tape used on post-mounted delineators shall be the same colour as the pavement markings they supplement.

The delineators should be either 15 cm x 15 cm (6" x 6") squares or 7.6 cm wide x 30 cm long (3" x 12") rectangles.

4.5.6 Vertical Panels

Vertical panels have a retroreflective, striped face that is at least 300 mm (12”) wide and 600 mm (24”) high. They may be stand mounted, fixed (bolted) directly to the road surface, or fixed by a proprietary curb mounting system.

They shall have alternating, diagonal fluorescent orange and white retroreflective stripes sloping downward at a 45-degree angle in the direction road users are to pass.

Where space is limited, vertical panels may be used to channelize vehicular traffic, divide opposing lanes, or replace barricades. On curves, they should be angled towards approaching traffic.

Vertical panels are not to be used for tapers in high speed environments, but may be used for tapers in urban low speed environments where the regular posted speed limit is ≤ 60 km/h.

When used as channelizing devices, vertical panels shall be secured such that the side facing traffic is at least 300 mm (12”) wide and 600 mm (24”) high.



Figure 4.12 A: Vertical Panel



Figure 4.12 B: Example of Curb-mounted Vertical Panels[1]

[1] Picture from http://www.qwickkurb.com/work-zone-applications/

4.5.7 Direction Indicator Barricade/Panel

Direction Indicator Barricades/Panels have a One-Direction Large Arrow sign mounted above a diagonal striped, horizontally aligned, retroreflective rail. The One-Direction Large Arrow sign shall be black on an orange background. The stripes on the bottom rail shall be alternating diagonal fluorescent orange and white retroreflective stripes sloping downward at a 45-degree angle in the direction road users are to pass.

The stripes are 100 mm (4") wide. The One-Direction Large Arrow sign is 60 cm x 30 cm (24" x 12"). The bottom rail shall have a length of 60 cm (24") and a height of 200 mm (8").

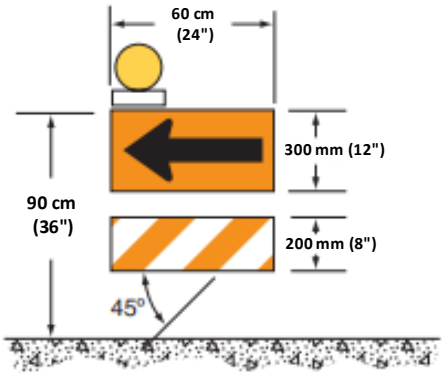


Figure 4.13 A: Direction Indicator Barricade/Panel

The Direction Indicator Barricade/Panel may be used in tapers, transitions, and other areas where specific directional guidance to drivers is necessary. Direction Indicator Barricades/Panels should be placed uniformly in series to direct the driver through the transition and into the intended travel lane. As an alternative to barrels in tapers/transitions, spacing patterns should coincide with that for barrels.

Due to the sign panels wind sail area, and overall light weight, direction indicator barricade/panels to prevent blow over on higher speed facilities may require weighting by methodologies such as sand bags, or other methods.



Figure 4.13 B: Direction Indicator Barricade/Panels Used for Taper



Figure 4.13 C: Example of Direction Indicator Barricade/Panels used with Vertical Panels[2]

[2] Picture from http://www.plasticade.com/traffic_safety/channelizers/verticade/

4.5.8 Barricades

Barricades are portable or fixed devices that are highly visible and relatively frangible. They are used to mark or restrict all or a portion of a roadway, especially areas into which most traffic is not to proceed, and are used in a series to channelize road users. Barricades are not designed to contain or redirect errant vehicles, and are not to be used to replace barriers.

There are three barricade types, each with particular applications. If appropriate, a higher type barricade may be substituted for a lower type.

Type 1 and Type 1A (formerly Class I and Class IA) barricades have one rail board. They are used on conventional roads and urban streets, generally for marking temporary hazards, delineating areas temporarily closed to traffic, and channelizing vehicles and pedestrians.



Figure 4.14 A: Type 1A Barricade



Figure 4.14 B: Type 1 Barricade

Type 2 (formerly Class II) barricades have two rail boards. They are used for temporary closures of high-volume, low-speed urban roads, and for channelization and temporary closures that will be in place for several days.



Figure 4.14 C: Type 2 Barricade

Type 3 (formerly Class III) barricades have three rail boards, and are used to close or partially close roads, and for temporary closures that will last for some time.

Type 2 and Type 3 barricades should be used on freeways, expressways, and other high-speed roads.

All barricade types can be used as sign supports.

Type 1A barricades shall be at least 60 cm (24") wide. Types 1, 2, and 3 barricades should be at least 1.2 metres (47") wide. Each rail board shall be 200 mm to 300 mm (8" to 12") wide.

Each rail shall have alternating fluorescent orange and white retroreflective stripes, sloping downward at a 45-degree angle. Rail stripe widths are 150 mm (6"). 100 mm (4") wide stripes may be used if rail lengths are less than 90 cm (36").

Alternating black and orange stripes are also acceptable, and may be a better option, depending on the background and contrast presented by the environment. An assessment of the area should be conducted to determine the most suitable colour pattern.



Figure 4.15: Type 1 Black and Orange Barricade

Follow these guidelines when using barricades:

1. Where barricades extend entirely across a roadway, the stripes should slope downward in the direction toward which road users are to pass. Where both right and left turns are provided, the stripes should slope downward in both directions from the center of the barricade or barricades. Where no turns are intended, the stripes should slope downward toward the center of the barricade or barricades.
2. Barricades used on expressways, freeways, and other high-speed roadways shall have at least 1,700 square centimeters (270 square inches) of retroreflective area facing traffic. Where traffic may approach a barricade from either side, the barricade should be retroreflective on both sides, or two barricades should be positioned back-to-back.
3. Road Closed R-012, Local Traffic Only R-012-T, and Detour C-005-LR signs may be attached to the highest barricade rail if required.
4. When a highway is closed but access is still allowed for local road users, barricades are not normally extended completely across the roadway.
5. Flashers may be used on top of the barricade to increase visibility (see *Section 4.9.2: Yellow Warning Lights*).
6. The stability of portable barricades can be enhanced by using sandbags, provided that they are placed on or close to the barricade bases. Weight should not be placed on the top of any rail. Non-deformable objects like rocks or concrete should not be used to weight the barricade.

4.5.9 Temporary Roadside/Median Barrier

In temporary traffic control, barrier and barricades are two different and distinct devices.

- **Barricades** (see *Section 4.5.8: Barricades*) are lightweight devices that are relatively forgiving of errant vehicles. They are normally placed at or nearly at right angles to approaching traffic to provide visual identification of hazardous locations and to delineate travel paths.
- **Barrier** is designed to contain and redirect errant vehicles. It is a solid, continuous installations designed to deflect errant vehicles at a small angle, thereby preventing them from entering a closed or hazardous area. It is normally placed parallel to or nearly parallel to approaching traffic.

Traffic Control Plans should include details regarding barrier installations. Barrier should be designed to meet American Association of State Highway and Transportation Officials (AASHTO) Manual for Assessing Safety Hardware (MASH) unless otherwise specified by the Road Authority.

The Ministry requires that temporary barrier, flares, and/or crash attenuators be installed in accordance with the latest edition of the BC Supplement to Transportation Association of Canada (TAC) Geometric Design Guide for Canadian Roads, or in accordance with manufacturer specifications (in the case of proprietary barriers) under the direction of an Engineer.



Figure 4.16: Concrete Roadside Barrier

These alternatives to barriers should be considered because of the risks to drivers and to the workers involved in installing and removing temporary traffic barriers:

- nightly backfill of excavations
- temporary tapers
- temporary detours or crossovers
- for lower-speed projects, additional or closer spacing of channelizing devices in conjunction with extra delineation (e.g., temporary raised pavement markers), and extra warning signs in advance of and within the work activity area

Temporary barrier is typically used:

- To separate road users from work activity areas, such as excavations, exposed objects, and material storage sites.
- To separate opposing directions of traffic.
- To separate workers, cyclists, and pedestrians from vehicle traffic.
- To separate traffic from drop-offs greater than 30 cm (12”) (see *Section 6.5: Treatment of Drop-Offs and Travel Lane Excavations*).

Follow these guidelines when using barrier:

- Barriers used as channelizing devices should be equipped with reflectors and/or Type A, B, or C yellow lights (see *Section 4.9.2: Yellow Warning Lights*).
- If sufficient room is available, a solid lane edge line may be installed to indicate shy distance.
- When barrier is used for lane closures, they shall be preceded by devices placed for a standard lane closure taper.
- When barrier restricts roadway width, enough width should be provided for the largest anticipated vehicle.

4.5.10 Temporary Lane Separators

Temporary lane separators may be used to:

- channelize road users
- divide opposing vehicular travel lanes
- divide lanes when two or more lanes are open in the same direction
- provide continuous pedestrian channelization

They may be supplemented by any of the other approved channelizing devices identified in *Section 4.5: Channelizing Devices*, such as tubular markers, vertical panels, and other devices used as lane dividers for opposing traffic.

Temporary lane separators should have retroreflectivity.

Temporary lane separators have a low-profile base designed for connecting the individual separator units together. The base shall be no more than 100 mm high and 300 mm wide, and have sloping sides to facilitate crossover by emergency vehicles.

Each unit should be temporarily affixed to the pavement.

At pedestrian crossing locations, temporary lane separators shall have an opening to provide a pathway that is at least 1.5 metres wide.

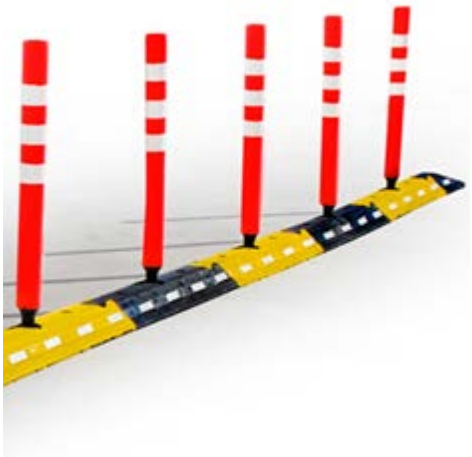


Figure 4.17: Temporary Lane Separators

4.5.11 Longitudinal Channelizing Devices

Longitudinal channelizing devices are lightweight, deformable devices which are highly visible and can be connected together to provide continuous delineation. They may be hollow, and may be filled with water as ballast only in areas where the water cannot freeze.



Figure 4.18: Interconnecting Device

Follow these guidelines when using longitudinal channelizing devices:

1. If used singly as Type 1, 2, or 3 barricades, longitudinal channelizing devices should comply with the general size, colour, stripe pattern, retroreflectivity, and placement standards for the barricades.
2. They may be used instead of a line of cones, tubes, or drums.
3. When used at night, they should include retroreflective material for improved visibility.
4. They may be used for pedestrian control, in which case they should be interlocked to channelize flow. The interlocking devices should not have gaps that allow pedestrians to stray from the channelizing path.
5. They need not meet the crash-worthy requirements for temporary traffic barriers so they should not be used to shield obstacles or to provide positive protection for pedestrians or workers.

4.5.12 Other Channelizing Devices

Channelizing devices other than the standard devices described in this Manual may be suggested for use in work zone applications. Using non-standard devices for work on roadways and rights-of-way requires the approval of the Road Authority.

Other channelizing devices should conform to the general size, colour, pattern, retroreflectivity, and placement standards described above.

4.6 Flashing Arrow Boards (FABs)

Flashing arrow boards (FABs) are signs with a matrix of elements that are capable of either flashing or sequential displays. They are very effective both day and night, providing additional warning and directional information that assists with controlling and merging road users through or around a work zone.

Their main purpose on multilane roadways is to direct traffic from a closed lane into another available lane with appropriate arrow indications. They can be used for either static or moving operations. Normally only one arrow head is displayed at a time.

Without directional indication, FABs can be used in place of, or in addition to, 4-way flashers and 360-degree warning lights to create a more visible warning that work is in progress. They can be mounted on trucks or trailers for both stationary and moving operations.

FABs **shall not be used** in directional display mode when:

- A lane closure is not required.
- All the work is on or outside the shoulder, and there is no need to close the adjacent travel lane.
- A Traffic Control Person is controlling traffic on what is normally a two-lane, two-way roadway.

4.6.1 FAB Specifications

1. Types

Flashing arrow boards are differentiated by size, where:

- **Type A** arrow boards are for low-speed urban streets.
- **Type B** arrow boards are for intermediate-speed facilities and maintenance or mobile operations on high-speed roadways.
- **Type C** arrow boards are for projects on high-speed, high-volume highways.
- **Type D** arrow boards are for use on vehicles authorized by the Road Authority.

Type A, B, and C arrow boards should have a solid rectangular appearance. Type D arrow boards conform to the shape of the arrow. The Ministry includes, under Type D, 16-lamp minimum arrow sticks with arrow heads. All arrow board faces (excluding arrow sticks) should be finished in non-reflective black.



Figure 4.19 A: Type D Arrow Board

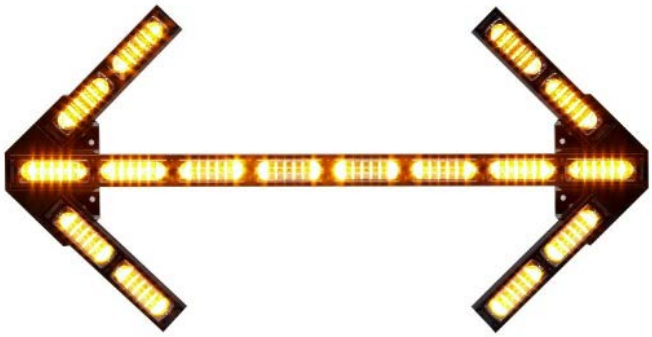


Figure 4.19 B: Example of Type D Arrow Board - Arrow Stick with Arrowheads[3]

[3] Picture from: https://www.whelen.com/auto/product?head_id=10&cat_id=70&prod_id=387

2. Minimum Requirements

FABs shall meet minimum requirements for size, legibility distance, number of elements, and other factors for the highway classification on which they are used.

Table 4.7: Minimum Requirements for Arrow Board Panels

Arrow Board Type	Minimum Size (arrow length x arrow-head width)	Minimum Visibility Distance	Minimum Number of Elements	Application (regular posted speed limit)
A rectangular board	120 cm x 60 cm (48" x 24")	600 metres	12	Short duration work ≤ 60 km/h
B rectangular board	152 cm x 75 cm (60" x 30")	800 metres	13	Short or Long duration work ≤ 60 km/h
C* rectangular board	244 cm x 120 cm (96" x 48")	1,000 metres	15	Short or Long duration work ≥ 70 km/h
D arrow-shaped board (truck-mounted)	120 cm x 60 cm (48" x 24")	600 metres	12 (16 for arrow shaped sticks with arrowheads)	Mobile work OR Short duration work ≤ 60 km/h

Notes:

The 90 cm x 45 cm size is no longer included but may continue to be used for lower speed applications (≤ 50 km/h).

* For mobile operations, truck mounted arrow shaped boards at least 150 cm x 75 cm may be used instead of Type C arrow boards.

3. Visibility and Light Intensity

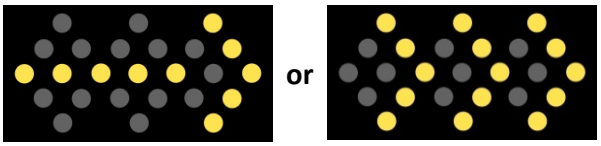
When using smaller arrow boards, ensure that the sign is conspicuous to approaching drivers and the arrow shape is retained, particularly for lane closures.

Vehicle-mounted arrow boards should have remote controls and elements capable of at least 50% dimming from full brilliance. Full brilliance should be used for day-time operations, and a dimmed mode should be used for night-time operations.

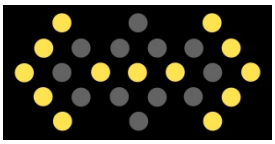
4. Display Mode

Yellow elements are used to display the three common display modes on FABs:

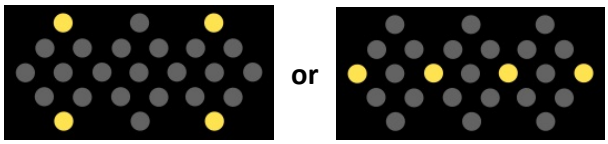
1. flashing arrow, sequential arrow, or sequential chevron directional display mode:



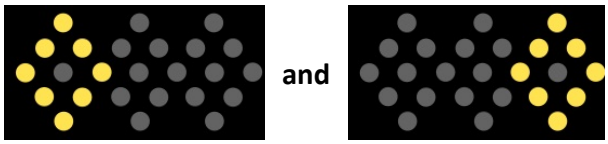
2. flashing double arrow mode:



3. flashing caution mode OR alternating diamond caution mode:



OR



The elements of a bulb-matrix FAB should be recess-mounted or equipped with an upper hood of not less than 180 degrees.

4.6.2 FAB Setup

A FAB is mounted on a vehicle, trailer, or other suitable support.

The minimum mounting height measured vertically from the bottom of the board to the roadway should be 2.0 m. Vehicle-mounted arrow boards are mounted at a height of at least 1.0 m.

A FAB should always be used in combination with appropriate signs, channelizing devices, and/or other temporary traffic control devices. It is generally placed within a closed lane. It should be delineated with tubes or drums at all times. When it is not being used, the FAB should be removed from the roadway.

When a FAB is placed on the shoulder in caution mode, it should be delineated with an appropriate shoulder closure taper.

For short-duration work on high-speed, non-freeway roadways, Type A FABs may be used on larger utility vehicles if Type B or Type C FABs cannot be physically accommodated.

Driver sight lines should be assessed when placing FABs to ensure maximum visibility without creating a hazard. Considering the curvature of the roadway, place the FAB in a position where there are no visual obstructions between it and the driver.

FABs should be set up as follows:

1. **For a lane closure that uses a stationary FAB (trailer-mounted),** the arrow board should be positioned at one of the following locations:
    a. On the shoulder (outside the travel lane), at the beginning of the merging taper, or
    b. Within the closed lane, at the end of the merging taper.
2. **For a lane closure that uses a mobile FAB (truck-mounted),** the arrow board should be positioned to provide enough separation from the work operation to allow approaching drivers to react appropriately.
3. **For multiple lane closures,** a separate arrow board shall be used for each closed lane.

4.6.3 FAB Operation

1. Determine the appropriate FAB display option based on the traffic control layout.
2. For flashing and sequencing arrow boards, the minimum element "on time" shall be 50% for the flashing mode, with equal intervals of 25% for each sequential phase. The flash rate should be between 25 and 40 flashes per minute.

   Sequencing arrow panels have several arrowheads that flash in a series, directing traffic to the right or left.
3. An arrow board in arrow or chevron mode can be used only for stationary or moving lane closures on multilane roadways.
4. An arrow board may be used in caution mode in situations that include, but are not limited to:
    - roadside work on or near the shoulder
    - temporarily closing one lane on a two-lane, two-way roadway
5. A Dynamic Message Sign (DMS) may be used to simulate an arrow board display.

4.6.4 Arrow Stick (Without Arrowhead)

Arrow sticks are vehicle-mounted sequential flashing devices used to supplement other temporary traffic control devices. They shall not be used as a replacement for FABs.

An arrow stick can be used to indicate "move/merge…right/left."

They can also be used as a flashing bar to indicate that caution is required.



Figure 4.20: Arrow Sticks

4.7 Automated Flagger Assistance Devices (AFADs)

An Automated Flagger Assistance Device (AFAD) is an automated flagging machine that features a circular red lens, a circular yellow lens, and a gate arm. It is used to stop traffic, but it is not a portable traffic signal (see *Section 4.8: Temporary and Portable Traffic Signals*). It is essentially an extension of the TCP's arm.

The TCP operates the AFAD using a remote control rather than a paddle to control traffic movement. This enables the TCP to be positioned outside the travel lane. Two AFADs can be operated by a single TCP at one end of the work activity or at a central location, or multiple AFADs can be operated by multiple TCPs, each positioned near an AFAD.



Figure 4.21: AFAD in STOP Position

4.7.1 Operation Modes

1. For road users to **stop**, the AFAD shall display a **steadily-illuminated red lens** with the gate arm in the down position.
2. For road users to **proceed**, the AFAD shall display a **flashing yellow** lens with the gate arm in the upright position.
3. For the **change** interval between flashing yellow and steady red, the AFAD shall display a **steadily-illuminated yellow lens** with the gate arm remaining in the upright position. The change interval should be at least 3 seconds unless a different duration is approved by engineering judgment. There is no change interval between the steady red and flashing yellow displays.

4.7.2 Deployment Guidelines

1. AFADs may be used only on two-lane, two-way roadways and on multilane roadways that have been reduced to one lane.
2. When used at night, the AFAD station shall be illuminated with overhead lighting.
3. The construction speed limit where AFADs are used shall be ≤ 70 km/h. High-speed roadways require a speed reduction.
4. An AFAD is not a traffic control signal, and it cannot be used to replace or substitute for a continuously-operating temporary traffic control signal.
5. An AFAD can be operated only by a TCP who has been trained to operate it.
6. A TCP operating an AFAD shall not leave it unattended at any time while it is in use.

4.7.3 Equipment Requirements

1. The AFAD shall have two 300 mm diameter signal lenses—i.e., a lens that displays solid red above a lens that displays flashing yellow. The flashing yellow lens shall also have solid yellow capability for change intervals.
2. The AFAD shall have a conflict monitor that prevents simultaneous illumination of the red and yellow lenses on the same device.
3. The AFAD shall have a gate arm with the following properties:
   - A fluorescent orange or red flag shall be installed at the end of the gate arm when the AFAD is in use.
   - The gate arm shall measure no less than 2.44 m (8 feet) in length and shall have a vertical aspect of at least 100 mm (4 in).• The gate arm shall lower and remain lowered on a red signal.
   - The gate arm shall rise to an upright position on a flashing yellow signal.
   - The gate arm shall have retroreflectivity on both sides with alternating fluorescent red and white bands. The bands shall be 200 mm (8”) long measured horizontally.
4. A black-on-white STOP HERE ON RED or STOP HERE ON RED SIGNAL sign shall be installed on the right side of the approach at the point where drivers are expected to stop. This sign is typically provided with the AFAD, and may be installed on it.
5. The AFAD shall be positioned in a location where the end of the arm shall reach at least to the center of the lane being controlled.

4.7.4 AFAD Placement and Operating Options

AFADs are placed either at each end of the work activity area or at one end of the work activity area with a TCP at the opposite end. Signing and AFAD placement are shown in *Section 7.9: Lane Closure with AFADs.*

The preferred operating option is to have a TCP controller for each AFAD. Assigning a TCP to each device becomes more critical on high-volume roadways and in more complex work zones where construction traffic may be entering and exiting frequently. For simpler, lower-volume situations where there are good sight lines, a single TCP may control up to two AFADs:

1. **Two-TCP Operation (Typical):**
   - One TCP operates each AFAD at either end of the work activity area; or
   - One TCP operates an AFAD at one end of the work activity area and the second TCP controls traffic with a paddle at the other end.

2. **Single-TCP Operation:**
   - One TCP positioned in a central location simultaneously operates two AFADs that are positioned at either end of the work activity area; or
   - One TCP operates a single AFAD that is positioned at one end of the work activity area while also controlling traffic with a paddle at the opposite end.

   In a single-TCP operation, all of these conditions shall be met:
   - The TCP has an unobstructed view of the AFAD(s).
   - The TCP has unobstructed views of approaching traffic in both directions.
   - The average daily traffic volume on the roadway is 6,000 vehicles or less.
   - The maximum distance between traffic control stations (TCP or AFAD) is 250 m.

Conflicting displays that release traffic in both directions simultaneously should be prevented by establishing clear communication procedures for fail-safe operation before work commences.

A TCP shall not activate the flashing yellow display (proceed) until the last vehicle from the opposing queue has cleared the work activity area.

4.8 Temporary and Portable Traffic Signals

Temporary traffic signals replicate Ministry traffic signals at intersections. They are a hard-wired traffic control system installed on standard Ministry signal poles or Ministry approved alternative, usually with an uninterruptible power supply (UPS), and are designed to Ministry standards. For more information on temporary traffic signals and their use, see the Ministry's Electrical and Traffic Engineering Manual or contact the Ministry's Traffic and Highway Safety Engineering team.

Portable traffic signals are mobile traffic control systems where a minimum of two signal heads are mounted on a self-contained trailer, usually powered by batteries whose charging mechanism is either through solar panels or a portable generator. These signals may include pedestrian and cycling type signal displays to address all types of road users.

Acceptance by the Road Authority is required prior to using Temporary or Portable Traffic Signals.

4.8.1 Portable Traffic Signals (Class 1 and 2) - Deployment Guidelines

A portable traffic signal is a mobile traffic control system which can control all types of road users. The portable traffic signal allows alternating directional flows where lane, shoulder and/or walkway constraints are in place.

Portable traffic signals consist of a minimum of two signal heads which are mounted on a self-contained trailer. These signals may include pedestrian and cycling type signal displays to address all types of road users. Both Class 1 and Class 2 portable traffic signals are capable of using various means, such as loops and/or push buttons, to activate displays. This allows on-demand alternative timing plans to be called based on road user type.

In addition, fixed-time portable traffic signals may have countdown timers which typically count down the time remaining until the signal turns green again. Countdown timers provide road users with knowledge of wait times. This reduces frustration and potential non-compliance to the signal display. Countdown timers are particularly advantageous in the following situations:

- When signals are far apart and clearance times are long,
- Where there is no visibility to the opposing signal, and/or
- When alternative timing plans are utilized for non-vehicular modes of traffic

Portable traffic signals can be used in specific circumstances to regulate single-lane alternating traffic during long-duration work—for example, on single-lane bridges and in rural construction environments. They may not be appropriate in mobile work zones and in work zones where there are several access and egress requirements for public and construction traffic.



Figure 4.22 A: Portable Traffic Signal Mounted on Trailer



Figure 4.22 B: Portable Traffic Signal with Countdown Timer – Red Display



Figure 4.22 C: Portable Traffic Signal with Countdown Timer – Green Display

Portable traffic signals should be inspected at least once a day as battery life is critical for operation. Other inspection frequencies may be used if justification is accepted by the Road Authority.

There are two operational classes of portable traffic signals:

1. **Class 1 Portable Signal:** A fixed-time signal used for short-duration work in low-speed environments (≤ 60 km/h) and where advance warning flashers are not required. A Traffic Engineer need not prepare the timing sheet for this signal.
2. **Class 2 Portable Signal:** An actuated signal or fixed-time signal used for long-duration work, and/or in high-speed environments (≥ 70 km/h), and/or where advance warning flashers are required. A Traffic Engineer shall prepare the timing sheet for this signal.

Portable traffic signals are used primarily to provide bi-directional traffic control in longer-term work zones. Typically, a pair of signals is set up at the perimeter of a roadway construction site, and signal communication is provided via radio interface.

Each signal unit shall have at least two signal heads for each approach and shall be positioned so that at least one signal head is overhead and one is side-mounted (see *Figure 4.22 A: Portable Traffic Signal Mounted on Trailer*). The signal heads should consist of three coloured displays with 300 mm (12") lenses.

See *Figure 7.10 Lane Closure with Temporary Signals* for details on site layout.

4.8.2 Portable Traffic Signals - Operational Guidelines

The operation of a portable traffic signal should consider:

- traffic volumes, including roadway and intersection capacity
- vehicle speeds
- work staging and operations
- sight distance restrictions
- affected side streets and driveways
- nature of adjacent land uses (e.g., residential or commercial)
- the use of pedestrian, cyclist, and/or custom user signal displays and/or audible signals
- signal phasing and timing requirements
- full-time or part-time operation
- actuated, fixed-time, or manual operation
- advance warning flashers
- power failures or other emergencies

The signal units should be powered by reliable power sources capable of operating the signals at all times unless traffic is controlled by Traffic Control Persons. The units may also be capable of communicating information remotely to traffic management personnel, such as errors or low battery levels.

Portable traffic signals shall be documented in the Traffic Control Plan and implemented in accordance with the standards specified in this Manual. Records shall be kept that identify placement, signal timing, inspection, and maintenance.

Drums should be placed on the approach side of the signals to provide notification and protection for road users, including cyclists.

Portable traffic signals that are not in use should be covered or removed.

Additional features to be included in a portable traffic signal system are:

- manual override to hold signal in green
- conflict monitor to ensure that the two signals in a pair cannot show green simultaneously
- ability to revert to flashing red mode if a fault is detected (i.e., low battery, lamp defect, lost communication, etc.)
- low-battery warning system, if applicable
- vehicle detection

Advance warning flashers are required where one or more of the following conditions apply (see also the Ministry's Electrical and Traffic Engineering Manual, Section 400):

- visibility of the signal is obstructed because of vertical or horizontal alignment
- grade approaching the signal requires more than normal braking effort
- regular posted speed limit for the highway is ≥ 70 km/h
- Road Authority has requested advance warning flashers

The Traffic Signal Ahead C-112 sign should be used in accordance with *Appendix B: Standard Construction Signs*.



C-112
Traffic Signal Ahead

Portable traffic signals should be inspected and logged at least once a day for:

- traffic operation (vehicle delay and throughput)
- signal alignment
- signal display failures
- power supply (battery life)
- signs of vandalism

If Class 2 portable signals will be used on a project, the Traffic Management Plan shall include:

- **Portable Traffic Signal Timing Plan** stamped by a Traffic Engineer.
- **Portable Traffic Signal Site Layout Plan** indicating the location of the site, position of the signals relative to the project site, distance between stop bars, distance between stop bars and advance warning sign (if applicable), location of vehicle-detection system, devices used to protect the signals, and other relevant information.

4.8.3 Portable Traffic Signal Timing Plan

When preparing the traffic signal timing sheets and supporting documentation for Class 2 traffic signals, refer to the Ministry's Electrical and Traffic Engineering Manual, Section 400. Signal timing calculations for simple fixed-time setups are provided below.

Table 4.8: Fixed Timing for Class 1 Portable Traffic Signals

Length of Single Lane (m)	15	30	45	60	75	90	105	120	135	150	165	180	195	210	225	240	255	270	285	300	315	330	345	360
"All Red" Interval One Way (s)	2	4	6	9	11	13	15	17	19	22	24	26	28	30	32	35	37	39	41	43	45	47	50	52
150	810	785	760	725	700	675	650	625	605	570	545	520	495	470	450	415	390	370	345	325	300	275	245	220
140	800	775	750	710	685	660	635	610	580	545	520	490	470	445	420	380	360	335	310	285	260	235	200	175
130	795	765	735	695	670	640	610	585	560	520	490	465	440	410	385	345	320	290	270	240	215	195		
120	785	755	725	680	650	620	590	560	535	490	460	430	405	375	345	305	275	250	220					
110	775	740	710	660	630	595	565	530	500	460	425	395	360	330	300	255								
100	760	725	690	640	600	570	535	500	465	415	380	345	315	285										
90	745	710	670	615	575	540	500	460	420	365	330	295												
85	740	700	660	600	560	520	480	440	395	340	305													
80	730	690	645	585	545	500	455	410	370	315														
75	725	680	630	565	520	475	430	385	340															
70	715	670	620	545	500	450	405	360																
65	705	650	600	520	470	420	370																	
60	695	635	580	495	440																			
55	675	610	555	470																				
50	660	595	530																					
45	640	570																						
40	625																							

CYCLE LENGTHS (seconds)

Vehicles / Hour on Heaviest Approach

Notes:

1. Assumed operating speed of 25 km/h through the work activity area.
2. Minimum Green Time approximately 15 seconds.
3. Yellow Clearance Interval of 3 seconds.
4. Based on 50% probability of queue clearance.

Example:

Given: Heaviest Approach Volume (One Way) = 365 veh/h

Length of Single Lane Section = 150 m

Find: Length of Green Interval (One Direction)

Length of All Red Intervals

Solution: By applying the given figures to the table above, we find that:

Cycle Length = 90 seconds

All Red Intervals = 22 seconds

Since the Green Time for each approach is equal to the Cycle Length minus two All Red Intervals (22 sec) minus two Yellow Clearance Intervals (3 sec), divided by two, then:

Green Time for each approach = $\frac{90 - (2 \times 22) - (2 \times 3)}{2}$ = 20 seconds

Class 2 Actuated Traffic Signal Timing Plan

PORTABLE TRAFFIC SIGNAL TIMING PLAN

Date:	Location:	Drawing:	Project:
	Portable Signal on Somewhere Road, Somewhere City	TE-00000-0	000/00000-1234

Phase Settings	Direction A	Clearance A	Direction B	Clearance B
	Somewhere Road E/B		Somewhere Road W/B	
Minimum Green	10.0	13.0	10.0	13.0
Maximum Green	35.0	13.0	35.0	13.0
Passage	5.0	-	5.0	-
Yellow	4.0	-	4.0	-
All Red Clearance	1.0	0.00	1.0	0.00
Intersection Flash	RED	-	RED	-
Advance Warning Time	5.0	-	5.0	-

COMMENTS:
Stop-bar to Stop-bar distance = 140 m
Assumed Clearance Speed = 40 km/h
Clearance Time for Direction A = 13 s
Clearance Time for Direction B = 13 s
Approach Speed = 60 km/h
Approach Grade in Direction A = 0%
Approach Grade in Direction B = 0%
Stop-bar to Advance Warning Sign Direction A = 58 m
Stop-bar to Advance Warning Sign Direction B = 58 m
Controller rests in RED when no calls exist

Example of Actuated Portable Traffic Signal Timing Calculations (Engineering Analysis Required)

Work Zone Characteristics:

Work zone length = 60 m
Approach speed = 60 km/h = 16.7 m/s
Assumed clearance speed = 40 km/h
Grade = 0% (both approach directions)

Stop bar distance from work zone (west side) = 40 m
Stop bar distance from work zone (east side) = 40 m

Therefore, stop bar to stop bar distance = 60 + 40 + 40 = 140 m

Portable Traffic Signal:

Select Max Green Time of 35 s — 35 s is often a good starting point for Max Green Time. However, based on traffic volumes and local knowledge, the Traffic Engineer may adjust the Max Green Time to ensure that there is no undue vehicle delay.

Clearance Time $R = \frac{3.6D}{V}$

$= \frac{3.6\ (140\ \text{m})}{40\ \text{km/h}}$

$= 13\ \text{s}$

Where: R = all Red Clearance Times (s)
V = travel speed through work zone (km/h)
D = distance between stop bars (m)

Yellow Time = 4 s

Passage Time of 5 s — 3 to 5 s is often a good starting point for Passage Time. However, based on traffic volumes and local knowledge, the Traffic Engineer may adjust the Passage Time.

Advance Warning Flashers:

Distance (stop bar to advance warning flasher) = 58 m for 60 km/h approach speed and 0 % grade.

Advance Warning Time $= \frac{D+D\rho}{V}$

$= \frac{58\ \text{m} + 21.3\ \text{m}}{16.7\ \text{m/s}}$

$= 5\ \text{s}$

Where: D = distance of flasher to signal (m)
Dρ = minimum perception distance = 21.3 m
V = posted speed limit (m/s)

Advance Warning Sign Distances

From the Electrical and Traffic Engineering Manual Section 402.6.10 Advance Warning Flashers:

		SIGN DISTANCE (m)					
		0.36[1]	0.34[1]	0.32[1]	0.31[1]	0.30[1]	0.30[1]
GRADE		50 km/h	60 km/h	70 km/h	80 km/h	90 km/h	100 km/h
D	-8%	49	71	100	132	170	207
O	-7%	48	70	97	127	164	199
W	-6%	47	67	94	123	158	192
N	-5%	46	66	91	119	153	185
H	-4%	45	64	88	116	148	179
I	-3%	44	62	86	112	143	173
L	-2%	43	61	84	109	139	168
L	-1%	42	60	82	106	135	163
	0%	41	58	80	104	131	159
U	+1%	41	57	78	101	128	155
P	+2%	40	56	76	99	125	151
H	+3%	39	55	75	96	122	147
I	+4%	39	54	73	94	119	143
L	+5%	38	53	72	92	116	140
L	+6%	37	52	70	90	114	137
	+7%	37	51	69	89	111	134
	+8%	36	50	68	87	109	131

[1] Friction factor